

RAG Knowledge Base Deployment

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1. Course Content

- Master the process and methods for local deployment, debugging, and testing of the RAG knowledge base.
- Master the methods for expanding the RAG knowledge base according to your own task scenarios.

[!TIP]

- The RAG knowledge base helps provide reference knowledge for general AI large models in vertical fields, avoiding the illusionary responses of AI large models and increasing the model's ability to respond to knowledge in vertical fields.
- The RAG knowledge base can help robots quickly expand their generalization capabilities in different task scenarios.

2. Start the Dify Service

- Connect to the car's infotainment system via VNC or SSH, and enter the following command in the terminal:

```
sh ~/bringup_dify.sh
```

```
jetson@yahboom: ~  
ROS_DOMAIN_ID: 62 | ROS: humble  
my_robot_type: M1 | my_lidar: c1 | my_camera: usb  
-----  
jetson@yahboom:~$ sh ~/bringup_dify.sh  
WARN[0000] The "DB_USERNAME" variable is not set. Defaulting to a blank string.  
WARN[0000] The "DB_DATABASE" variable is not set. Defaulting to a blank string.  
WARN[0000] The "DB_PASSWORD" variable is not set. Defaulting to a blank string.  
WARN[0000] The "DB_PASSWORD" variable is not set. Defaulting to a blank string.  
WARN[0000] The "DB_DATABASE" variable is not set. Defaulting to a blank string.  
WARN[0000] The "DB_USERNAME" variable is not set. Defaulting to a blank string.  
WARN[0000] The "CERTBOT_EMAIL" variable is not set. Defaulting to a blank string  
.  
WARN[0000] The "CERTBOT_DOMAIN" variable is not set. Defaulting to a blank string.  
g.  
WARN[0000] The "DB_DATABASE" variable is not set. Defaulting to a blank string.  
WARN[0000] The "DB_PASSWORD" variable is not set. Defaulting to a blank string.  
WARN[0000] The "DB_USERNAME" variable is not set. Defaulting to a blank string.  
WARN[0000] The "DB_USERNAME" variable is not set. Defaulting to a blank string.  
WARN[0000] The "DB_DATABASE" variable is not set. Defaulting to a blank string.  
WARN[0000] The "DB_PASSWORD" variable is not set. Defaulting to a blank string.  
WARN[0000] The "DB_DATABASE" variable is not set. Defaulting to a blank string.  
WARN[0000] The "DB_PASSWORD" variable is not set. Defaulting to a blank string.  
WARN[0000] The "DB_DATABASE" variable is not set. Defaulting to a blank string.  
WARN[0000] The "DB_PASSWORD" variable is not set. Defaulting to a blank string.  
WARN[0000] The "DB_USERNAME" variable is not set. Defaulting to a blank string.  
[+] Running 10/10  
✓ Container docker-sandbox-1      Runni...      0.0s  
✓ Container docker-web-1          Running       0.0s  
✓ Container docker-ssrf_proxy-1   Ru...        0.0s  
✓ Container docker-db-1           Healthy       0.5s  
✓ Container docker-redis-1        Running       0.0s  
✓ Container docker-worker_beat-1   R...         0.0s  
✓ Container docker-plugin_daemon-1 Running       0.0s  
✓ Container docker-api-1          Running       0.0s  
✓ Container docker-worker-1        Runnin...    0.0s  
✓ Container docker-nginx-1        Running       0.0s  
jetson@yahboom:~$
```

- To check the car's IP address, you can use the OLED screen, `ifconfig`, or directly on the terminal. Simply enter the car's IP address into your browser's address bar to access the Dify management page.

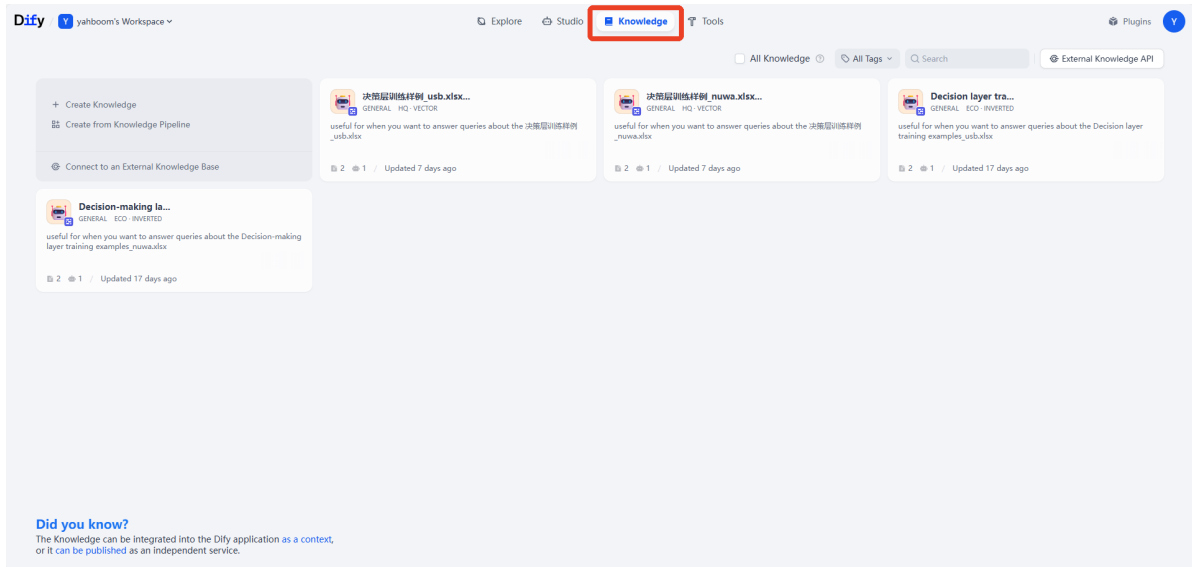
```
jetson@yahboom: ~  
jetson@yahboom: ~80x24  
[System Information]  
-----  
IP_Address_1: 192.168.2.49  
IP_Address_2: 192.168.12.34  
MACHINE: OrinNx | ROS_DISTRO: humble | ROS_DOMAIN_ID: 30  
ROBOT_TYPE: ROSMASTER-M3Pro | CAMERA_TYPE: dabai_dcw2 | RADAR: Tmini-plus*2  
-----  
jetson@yahboom:~$
```

3. View the Preset Knowledge Base

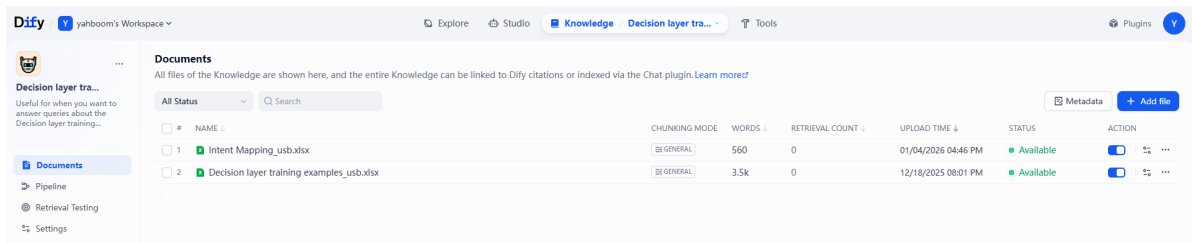
- Click on the Knowledge Base page on the homepage. Dify comes with two pre-set RAG knowledge bases with identical content, only differing in language.

[!TIP]

- The pre-set knowledge bases provide training examples for various task scenarios to help AI models quickly master relevant skills.

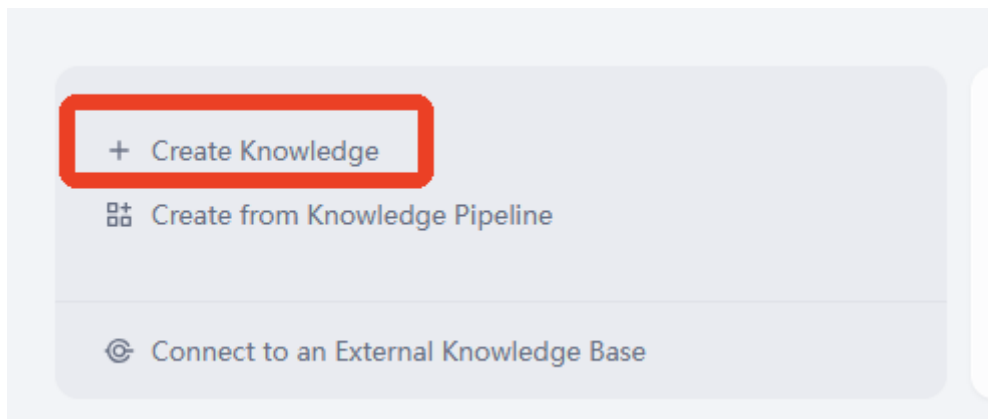


- Open a knowledge base, which contains two preset files: `Intent mapping_usb_a1.xlsx` and `Decision layer training examples_usb.xlsx`. Where:
- Intent Mapping: Stores user intents and the corresponding tasks the robot needs to perform.
- Decision Layer Example Library: Stores preset task scenario reference examples.



4. Expanding the RAG Knowledge Base

- If you need to expand the knowledge base with a new one, click Create Knowledge.



- Here, we'll use imported local data as an example.
- Click Import from file —> Browse —> Next

Data Source

Import from file Sync from Notion Sync from website

Upload file

Drag and drop file or folder, or **Browse**

Supports XLS, VTT, DOCX, MDX, CSV, PROPERTIES, HTML, MARKDOWN, HTM, TXT, XLSX, PDF, MD. Max 5 in a batch and 15 MB each.

Decision-making layer training examples nuwa.xlsx
XLSX · 0.01MB

Next →

[I want to create an empty Knowledge](#)

- Afterwards, you'll enter the knowledge base configuration page. Clicking "Preview Block" will show the file's block segmentation effect. Here, we'll select "Economic" as the indexing mode.

[!TIP]

- For beginners, we recommend using the Economic mode for learning and testing first. The differences between the two indexing modes are:
- Economic: Retrieves content from the knowledge base using **keywords**. It cannot perform extended searches based on similar semantics, and the way it retrieves knowledge fragments is relatively rigid.
- High-Quality Mode: Requires embedding a model, consuming additional tokens, and requires re-ranking the model to achieve more accurate retrieval of similar semantic fragments.

4.1 Economic Mode Knowledge

- After selecting the following configuration, click Save and Process

Dify yahooboom's Workspace

Explore Studio **Knowledge** Tools Plugins

KNOWLEDGE 1024 characters 50 characters

Text Pre-processing Rules

☒ Replace consecutive spaces, newlines and tabs

☐ Delete all URLs and email addresses

☐ Chunk using Q&A format in English

Preview Chunk Reset

Parent-child

When using the parent-child mode, the child-chunk is used for retrieval and the parent-chunk is used for recall as context.

Index Method

High Quality (RECOMMEND)

Calling the embedding model to process documents for more precise retrieval helps LLM generate high-quality answers.

Economical

Using 10 keywords per chunk for retrieval, no tokens are consumed at the expense of reduced retrieval accuracy.

Retrieval Setting

Learn more about retrieval method, you can change this at any time in the Knowledge settings.

Inverted Index

Inverted Index is a structure used for efficient retrieval. Organized by terms, each term points to documents or web pages containing it.

Top K

3

Save & Process

PREVIEW Decision-making layer training examples nuwa.xlsx 14 ESTIMATED CHUNKS

Chunk-1 · 127 characters
query: "Track the yellow square"; answer: "1. Call the function to track the specified color square, the parameter is 'yellow'"

Chunk-2 · 329 characters
query: "Identify the object in my hand and follow it"; answer: "1. Call the visual function to obtain the coordinates of the upper left and lower right corners of the object in the picture 2. Record the coordinates of the object in the image as x1, y1, x2, y2 3. Call the object following function with parameters x1, y1, x2, y2"

Chunk-3 · 96 characters
query: "Start machine code following"; answer: "1. Call the machine code to follow the function"

Chunk-4 · 76 characters
query: "Start face following"; answer: "1. Call the face following function"

Chunk-5 · 94 characters
query: "Start human posture following"; answer: "1. Call the human posture following function"

Chunk-6 · 135 characters
query: "Start following the red square"; answer: "1. Call the function that follows the specified color square, the parameter is 'red'"

Chunk-7 · 118 characters
query: "Start patrolling the green line"; answer: "1. Call the tracking autopilot function with the parameter 'green'"

Chunk-8 · 238 characters
query: "Please go to the tea room and see what's around."; answer: "1. Call the function to record the current location and record the location 2. Navigate to the kitchen 3. Call the visual function to observe the surrounding environment"

Chunk-9 · 504 characters
query: "Navigate to the kitchen and observe the surroundings. If you see a colored line on the ground, start patrolling the line. If you don't find any, return to the starting point."; answer: "1. Call the function to record the current position and record the position 2. Navigate to the kitchen 3. Call the visual function to observe the surrounding environment 4. If there is a colored line"

- Afterwards, wait for the embedding to complete, then click "Go to Documentation"

 **Knowledge created**
We automatically named the Knowledge, you can modify it at any time.

 **Knowledge name**
Decision-making la...

EMBEDDING PROCESSING...
Decision-making layer training examples_nuwa.xlsx 0%

Chunking Setting: Custom

Maximum Chunk Length: 1024

Text Preprocessing Rules: Replace consecutive spaces, newlines and tabs

Index Method: Economical

Retrieval Setting: Inverted Index

Access the API

Go to document →

- When the knowledge base is functioning normally, its status will show as "Available." Then, click on the knowledge base file.

Documents							
All files of the Knowledge are shown here, and the entire Knowledge can be linked to Dify citations or Indexed via the Chat plugin. Learn more?							
All Status		Q Search		Metadata		+ Add file	
#	NAME	CHUNKING MODE	WORDS	RETRIEVAL COUNT	UPLOAD TIME	STATUS	ACTION
1	Decision-making layer training examples_nuwa.xlsx	GENERAL	3.6k	0	12/31/2025 02:20 PM	Available	  

- Afterwards, we can see the segmented knowledge base fragments. The small text below each segment is the automatically generated keyword for that segment (only available in the economic mode).

Decision-making layer training examples_nuwa.xlsx  

14 CHUNKS

All Q Search

Chunk-01 · 127 characters · 0 Retrieval count

query: "Track the yellow square"; answer: "1. Call the function to track the specified color square, the parameter is 'yellow'"

#yellow #track #specifi... #answer #query #color #Track #Call #function #square

Chunk-02 · 329 characters · 0 Retrieval count

query: "Identify the object in my hand and follow it"; answer: "1. Call the visual function to obtain the coordinates of the upper left and lower right corners of the object in the picture 2. Record the coordinates of the object in the image as x1, y1, x2, y2 3. Call the object following function with parameters x1, y1, x2, y2..."

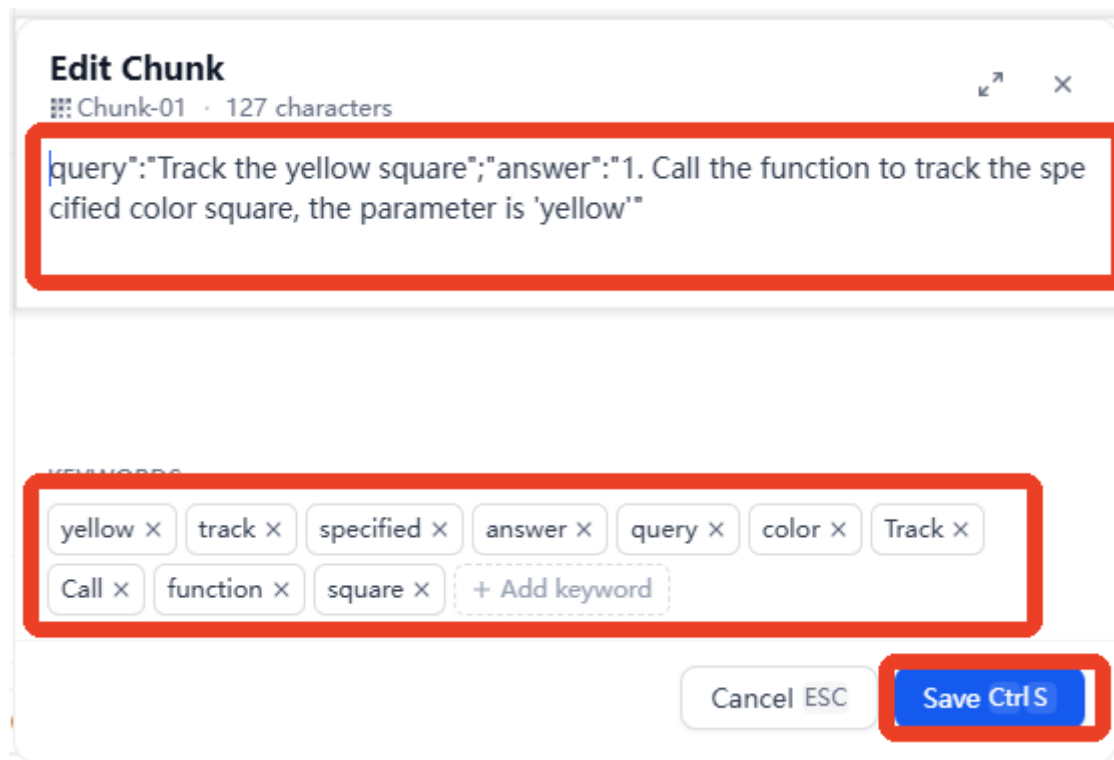
#x1 #Identify #function #y2 #object #y1 #coordi... #query #x2 #Call

Chunk-03 · 96 characters · 0 Retrieval count

query: "Start machine code following"; answer: "1. Call the machine code to follow the function"

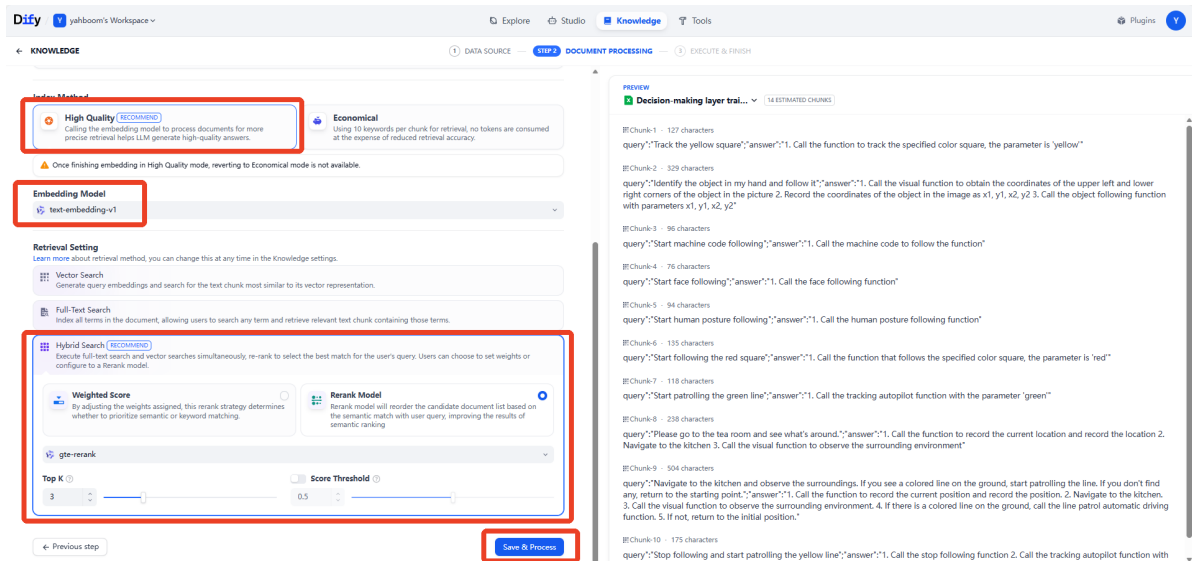
#code #follow #machi... #answer #query #Call #follow... #Start #function

- If the keywords do not accurately describe the knowledge fragment, click "Edit" on the right side of the fragment to edit the fragment content or keywords individually. The image below shows the keywords modified, then click "Save".



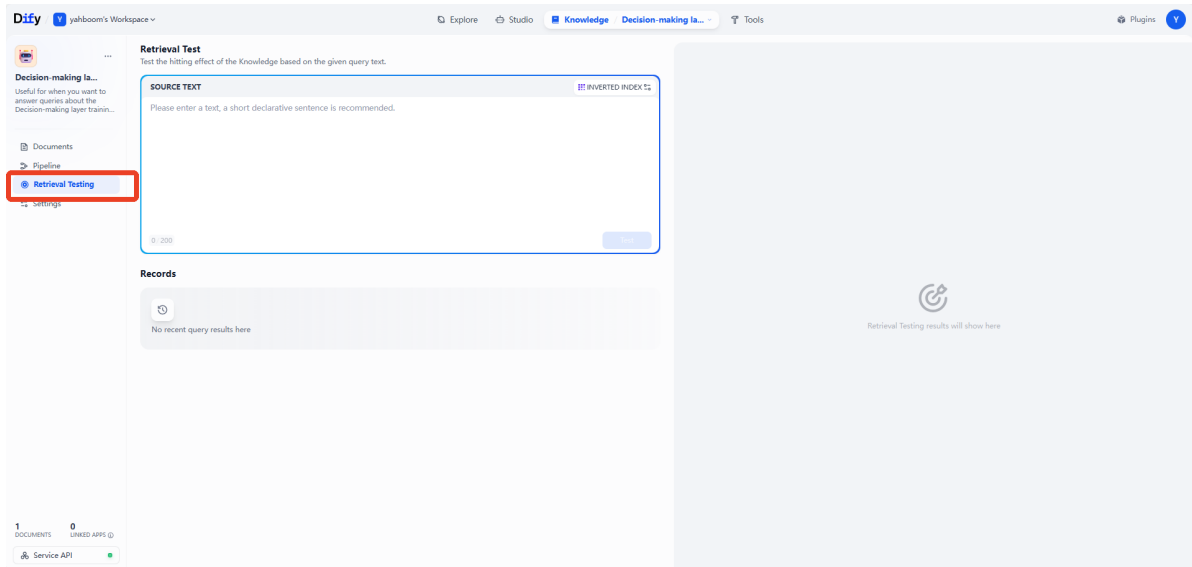
4.2 High-Quality Knowledge

- If you need to use a high-quality knowledge base later, refer to the tutorial in this section.
- The knowledge base creation and import file process is the same as before.
- Here, select high-quality indexing method and choose any retrieval method. We'll use mixed retrieval as an example here. Finally, save and process.



4.3 Retrieval Testing

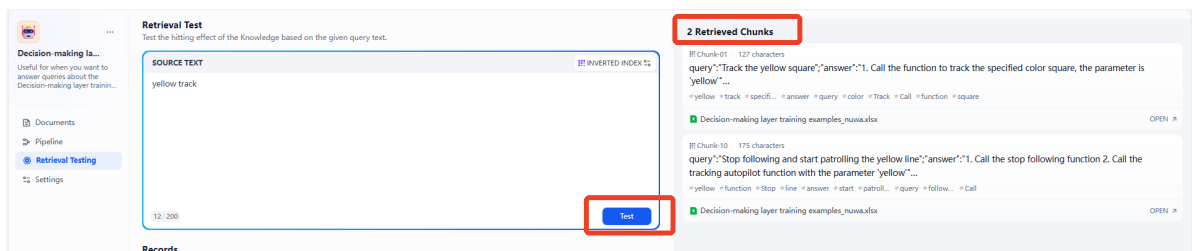
- Recall testing tests the actual effect of recalling relevant knowledge fragments from the knowledge base based on the input, helping to optimize the AI model's response performance.
- After opening a knowledge base, click "Recall Testing" on the left.



- Enter test content (simulating user input in actual use) in the source text, then click Test.
- The right side will display the recalled segments and the knowledge base related to the input content. Here, the test knowledge base is the Economic Model knowledge base, which is searched based on keywords.

[TIP]

- The default reference knowledge base is the Economic Model. If the recalled segments are not performing well when creating your own extended knowledge base, refer to [4.1 Economic Model Knowledge](#) to adjust the keywords for the corresponding segments.



- If it's a high-quality mode knowledge base, the recalled segments will have a **SCORE** score. The higher the score, the higher the match between the segment and the input content. The high-quality mode knowledge base can perform associative searches based on similar semantics, but it also requires tokens.

